

Multi-Agent Financial Research System

Red Flag Agent Design & Logical Architecture

1. Overview of the Financial Red Flag Agent

The **Financial Red Flag Agent (Agent #2)** is responsible for automatically identifying financial risks, anomalies, warning signals, and governance issues from uploaded financial documents such as annual reports, quarterly reports, 10-K filings, 10-Q filings, management discussion reports, auditor reports, cash flow statements, balance sheets, income statements, investor presentations, and other financial disclosures.

The agent analyzes structured financial data produced by the Financial Extraction Agent together with textual information extracted from the uploaded documents. Using rule-based validation, financial ratio analysis, historical trend analysis, and Large Language Models (LLMs), the agent detects potential financial red flags that may indicate business risks, accounting concerns, operational weaknesses, liquidity problems, or corporate governance issues.

The identified risks are transformed into a standardized JSON structure containing detected red flags, severity levels, supporting evidence, confidence scores, and recommended explanations. This structured output serves as valuable input for downstream agents such as the Research Agent, Comparison Agent, Report Generation Agent, and Visualization Agent.

The Financial Red Flag Agent combines financial analytics, intelligent document understanding, AI-powered reasoning, and explainable risk detection to ensure reliable identification of financial warning signals while maintaining complete traceability to the original financial documents.

2. Objectives

The Financial Red Flag Agent is designed with the following objectives:

- Automatically detect financial risks from uploaded financial documents.
- Identify liquidity, profitability, solvency, operational, and governance-related warning signals.
- Analyze financial statements together with qualitative disclosures.
- Detect abnormal financial trends and inconsistencies.
- Evaluate auditor observations and management risk disclosures.
- Assign severity levels for every detected financial risk.
- Generate confidence scores for identified red flags.
- Provide explainable risk insights for downstream AI agents.

3. Logical Architecture & Workflow

The Financial Red Flag Agent follows a structured workflow for analyzing every uploaded financial document.

Step 1: Receive Structured Financial Data

The agent receives structured financial metrics and document metadata generated by the Financial Extraction Agent.

Step 2: Load Financial Document Context

Relevant document sections including Management Discussion & Analysis (MD&A), Risk Factors, Auditor Reports, Notes to Financial Statements, and Corporate Governance disclosures are retrieved for further analysis.

Step 3: Financial Ratio Evaluation

The agent calculates and evaluates financial indicators such as liquidity ratios, debt ratios, profitability margins, cash flow adequacy, and leverage indicators to identify abnormal financial conditions.

Step 4: Rule-Based Risk Detection

Predefined financial rules are applied to detect warning signals including declining revenues, increasing debt, negative operating cash flow, recurring losses, inventory anomalies, and deteriorating financial health.

Step 5: AI-Based Risk Analysis

The processed financial information and supporting document context are provided to the Language Model, which identifies additional qualitative risks including legal issues, litigation, governance concerns, auditor warnings, regulatory compliance issues, and management uncertainties.

Step 6: Risk Classification

Detected financial issues are categorized into predefined risk categories such as Liquidity Risk, Credit Risk, Operational Risk, Market Risk, Governance Risk, and Financial Reporting Risk.

Step 7: Severity & Confidence Scoring

Each detected red flag is assigned a severity level (Low, Medium, High, Critical) together with confidence scores based on supporting evidence.

Step 8: Structured JSON Generation

The final financial risk analysis is returned as a standardized JSON containing detected red flags, explanations, evidence, severity levels, confidence scores, and source references for downstream processing.

4. Red Flag Detection Strategy

To ensure reliable and explainable financial risk detection, the Financial Red Flag Agent follows a strict analysis strategy.

System Instructions

- Detect only risks supported by the uploaded financial documents.
- Do not assume or invent financial issues.
- Preserve original financial context and supporting evidence.
- Validate numerical inconsistencies before generating warnings.
- Classify risks according to predefined categories.
- Return **null** whenever sufficient evidence is unavailable.
- Assign confidence scores for every detected red flag.
- Preserve supporting source text for explainability.
- Maintain consistency across all financial analyses.

This approach minimizes hallucinations while ensuring trustworthy and explainable financial risk assessment.

5. Red Flag Detection Pipeline

The Financial Red Flag Agent implements the following analysis workflow:

1. Receive structured financial data from the Extraction Agent.
2. Retrieve supporting document sections.
3. Validate extracted financial metrics.
4. Calculate financial ratios and trend indicators.
5. Apply rule-based financial risk detection.
6. Perform AI-powered qualitative risk analysis.
7. Classify detected risks into predefined categories.

8. Assign severity levels.
9. Calculate confidence scores.
10. Attach supporting evidence and document references.
11. Generate standardized JSON output.

6. Key Features

The proposed Financial Red Flag Agent provides several important features:

- Automated financial risk detection
- Rule-based anomaly identification
- AI-powered qualitative risk analysis
- Financial ratio evaluation
- Trend and variance analysis
- Auditor report analysis
- Governance issue detection
- Liquidity and solvency risk identification
- Confidence score generation
- Severity classification
- Explainable AI with supporting evidence
- Structured JSON output
- Error handling and validation
- Scalable enterprise-ready architecture

7. Advantages

The proposed architecture offers several advantages:

- Automates identification of financial warning signals.
- Reduces manual financial risk assessment effort.
- Combines quantitative and qualitative financial analysis.
- Improves decision-making through explainable AI.
- Detects risks across multiple financial document types.

- Provides confidence-based validation for detected issues.
- Enables seamless integration with downstream AI agents.
- Produces standardized structured financial risk reports.
- Supports enterprise-scale financial analysis workflows.

8. Conclusion

The Financial Red Flag Agent provides a scalable and intelligent solution for automated financial risk assessment. By combining structured financial analysis, rule-based validation, trend evaluation, AI-powered reasoning, and confidence-driven risk scoring, the system transforms financial documents into actionable financial risk insights.

This milestone establishes the logical foundation for automated financial risk detection within the Multi-Agent Financial Research System and enables subsequent agents to perform advanced research, company comparison, investment analysis, report generation, and financial decision support using explainable and evidence-backed financial risk assessments.